SE LAB REPORT – LAB 5

SRN - PES2UG24CS453
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Gitlink where every lab report doc is present - https://github.com/Priyans00/SE_lab_Sec_H_PES2UG24CS453
Gitlink for lab5 -> https://github.com/Priyans00/PetShelter_registrations

Static code analysis

Before ->

Bandit report -

Before_Reports > bandit_report.txt

```
Run started:2026-10-14 06:42:19.816851+00:00

Test results:
>> Issue: [B608:hardcoded_sql_expressions] Possible SQL injection vector through string-based query construction.
   Severity: Medium   Confidence: Medium
   CWE: CWE-89 (https://cwe.mitre.org/data/definitions/89.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b608_hardcoded_sql_expressions.html
   Location: .\pet_shelter.py:10:17
9   def registerPet(name, species, age, tags=[]):
10      conn.execute("INSERT INTO pets (name, species, age, adopted) VALUES ('%s', '%s', %d, 0)" % (name, species, age))
11      tags.append(species)

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\pet_shelter.py:22:24
21  def adoptionCode():
22      return "PET-" + str(random.randint(1000, 9999))
23

--------------------------------------------------
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\pet_shelter.py:36:4
35          conn.execute("DELETE FROM pets WHERE adopted = 0")
36      except Exception:
37          pass
38

--------------------------------------------------

Code scanned:
    Total lines of code: 37
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 2
        Medium: 1
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 1
        High: 2
Files skipped (0):
```

Flake 8 ->

Before_Reports > flake8_report.txt

```
pet_shelter.py:5:80: E501 line too long (113 > 79 characters)
pet_shelter.py:10:80: E501 line too long (116 > 79 characters)
pet_shelter.py:27:80: E501 line too long (103 > 79 characters)
pet_shelter.py:29:5: E722 do not use bare 'except'
pet_shelter.py:41:26: E701 multiple statements on one line (colon)

```

Pylint Report ->

Before_Reports > pylint_report.txt

```
************* Module pet_shelter
pet_shelter.py:5:0: C0301: Line too long (113/100) (line-too-long)
pet_shelter.py:10:0: C0301: Line too long (116/100) (line-too-long)
pet_shelter.py:27:0: C0301: Line too long (103/100) (line-too-long)
pet_shelter.py:1:0: C0114: Missing module docstring (missing-module-docstring)
pet_shelter.py:6:0: C0103: Constant name "adoption_count" doesn't conform to UPPER_CASE naming style (invalid-name)
pet_shelter.py:9:0: C0116: Missing function or method docstring (missing-function-docstring)
pet_shelter.py:9:0: C0103: Function name "registerPet" doesn't conform to snake_case naming style (invalid-name)
pet_shelter.py:9:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
pet_shelter.py:10:17: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
pet_shelter.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
pet_shelter.py:16:4: W0603: Using the global statement (global-statement)
pet_shelter.py:21:0: C0116: Missing function or method docstring (missing-function-docstring)
pet_shelter.py:21:0: C0103: Function name "adoptionCode" doesn't conform to snake_case naming style (invalid-name)
pet_shelter.py:25:0: C0116: Missing function or method docstring (missing-function-docstring)
pet_shelter.py:25:0: C0103: Function name "oldestAvailable" doesn't conform to snake_case naming style (invalid-name)
pet_shelter.py:29:4: W0702: No exception type(s) specified (bare-except)
pet_shelter.py:33:0: C0116: Missing function or method docstring (missing-function-docstring)
pet_shelter.py:33:0: C0103: Function name "clearUnadopted" doesn't conform to snake_case naming style (invalid-name)
pet_shelter.py:36:11: W0718: Catching too general exception Exception (broad-exception-caught)
pet_shelter.py:40:0: C0116: Missing function or method docstring (missing-function-docstring)
pet_shelter.py:40:0: C0103: Function name "shelterStatus" doesn't conform to snake_case naming style (invalid-name)
pet_shelter.py:41:27: C0321: More than one statement on a single line (multiple-statements)
pet_shelter.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)

-------------------------------------------------------------------
Your code has been rated at 3.95/10 (previous run: 3.95/10, +0.00)
```

After ->

Bandit Report ->

After_Reports > bandit_report.txt

```
Run started:2026-10-14 06:49:31.225408+00:00

Test results:
    No issues identified.

Code scanned:
    Total lines of code: 49
    Total lines skipped (#nosec): 0
    Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
    Total issues (by severity):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
    Total issues (by confidence):
        Undefined: 0
        Low: 0
        Medium: 0
        High: 0
Files skipped (0):
```

Flake8 Report ->

After_Reports > flake8_report.txt

```
Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing
```

Problems Output Debug Console Terminal Ports

```
(base) PS E:\code\se\petshelter> flake8 pet_shelter.py > flake8_report.txt
(base) PS E:\code\se\petshelter>
```

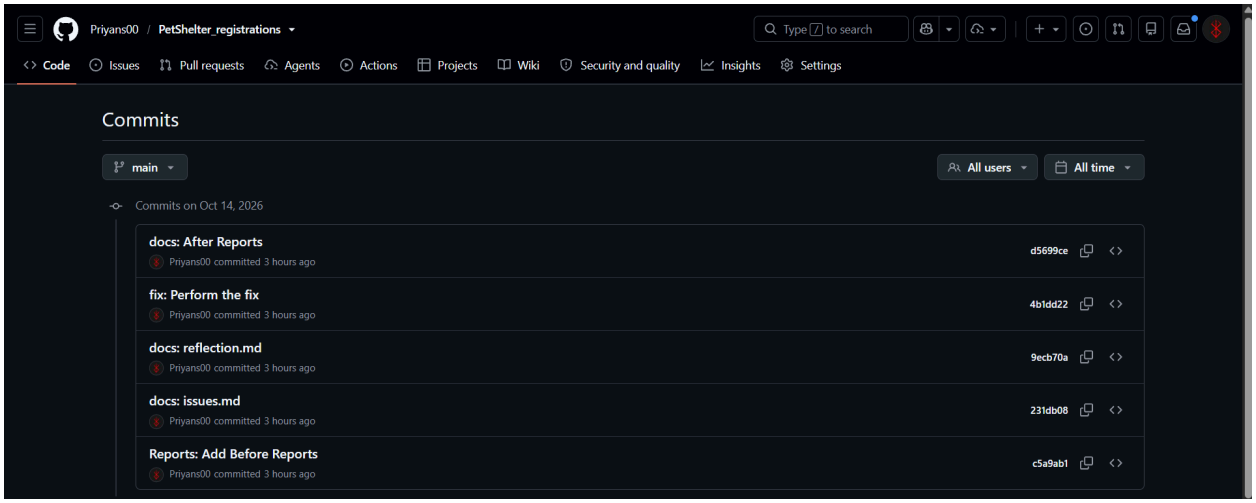
(Empty Report)

Pylint Report ->

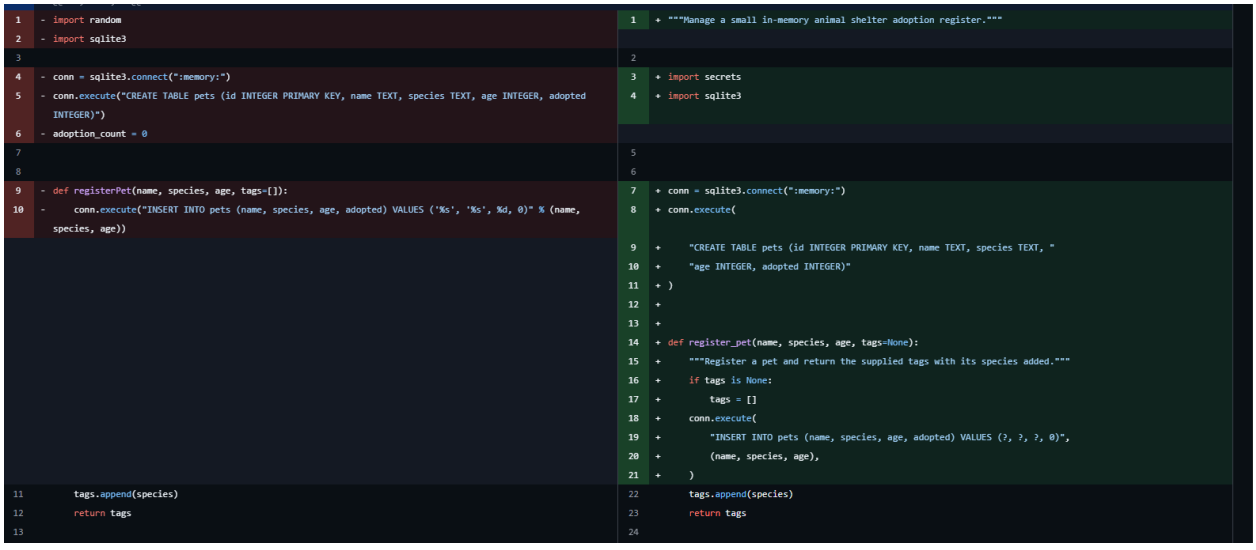
After_Reports > pylint_report.txt

```
1
2  ------------------------------------------------------------------
3  Your code has been rated at 10.00/10 (previous run: 3.95/10, +6.05)
4
5
```

Commits ->



Fixes ->



```
- import random
- import sqlite3

- conn = sqlite3.connect(":memory:")
- conn.execute("CREATE TABLE pets (id INTEGER PRIMARY KEY, name TEXT, species TEXT, age INTEGER, adopted INTEGER)")
- adoption_count = 0

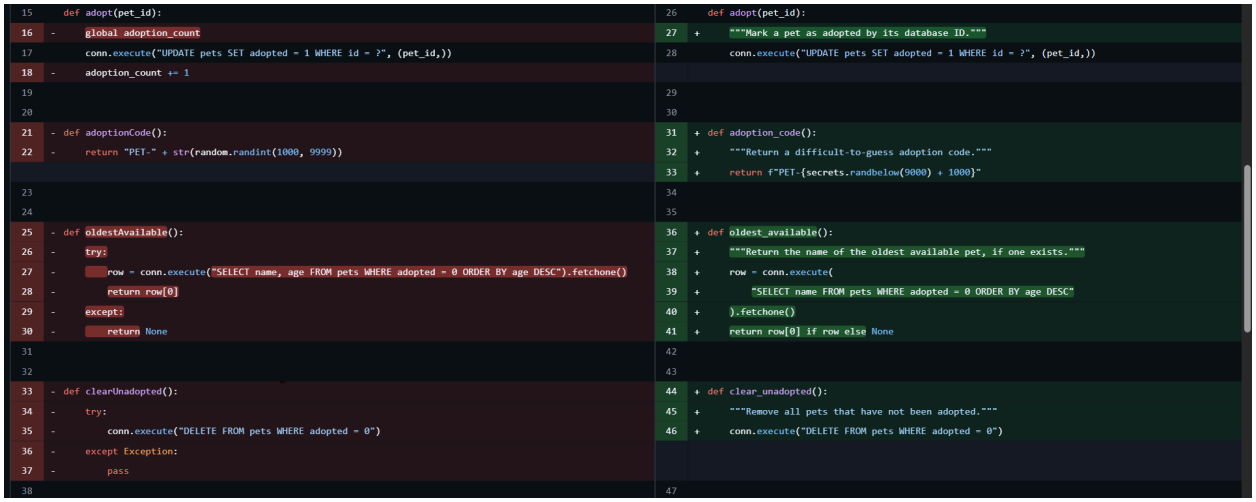
- def registerPet(name, species, age, tags=[]):
-     conn.execute("INSERT INTO pets (name, species, age, adopted) VALUES ('%s', '%s', %d, 0)" % (name, species, age))
      tags.append(species)
      return tags
```

```
+ """Manage a small in-memory animal shelter adoption register."""

+ import secrets
+ import sqlite3

+ conn = sqlite3.connect(":memory:")
+ conn.execute(
+     "CREATE TABLE pets (id INTEGER PRIMARY KEY, name TEXT, species TEXT, "
+     "age INTEGER, adopted INTEGER)"
+ )
+
+
+ def register_pet(name, species, age, tags=None):
+     """Register a pet and return the supplied tags with its species added."""
+     if tags is None:
+         tags = []
+     conn.execute(
+         "INSERT INTO pets (name, species, age, adopted) VALUES (?, ?, ?, 0)",
+         (name, species, age),
+     )
      tags.append(species)
      return tags
```

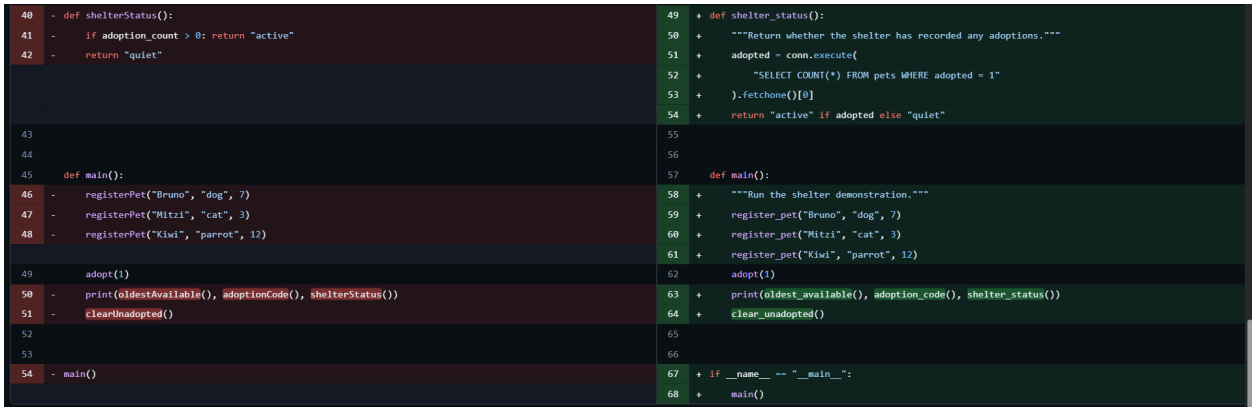
- Handling execution when tags is None
- Intendation fix



```
15    def adopt(pet_id):
16  -     global adoption_count
17        conn.execute("UPDATE pets SET adopted = 1 WHERE id = ?", (pet_id,))
18  -     adoption_count += 1
19
20
21  - def adoptionCode():
22  -     return "PET-" + str(random.randint(1000, 9999))
23
24
25  - def oldestAvailable():
26  -     try:
27  -         row = conn.execute("SELECT name, age FROM pets WHERE adopted = 0 ORDER BY age DESC").fetchone()
28  -         return row[0]
29  -     except:
30  -         return None
31
32
33  - def clearUnadopted():
34  -     try:
35  -         conn.execute("DELETE FROM pets WHERE adopted = 0")
36  -     except Exception:
37  -         pass
38
```

```
26    def adopt(pet_id):
27  +     """Mark a pet as adopted by its database ID."""
28        conn.execute("UPDATE pets SET adopted = 1 WHERE id = ?", (pet_id,))
29
30
31  + def adoption_code():
32  +     """Return a difficult-to-guess adoption code."""
33  +     return f"PET-{secrets.randbelow(9000) + 1000}"
34
35
36  + def oldest_available():
37  +     """Return the name of the oldest available pet, if one exists."""
38  +     row = conn.execute(
39  +         "SELECT name FROM pets WHERE adopted = 0 ORDER BY age DESC"
40  +     ).fetchone()
41  +     return row[0] if row else None
42
43
44  + def clear_unadopted():
45  +     """Remove all pets that have not been adopted."""
46  +     conn.execute("DELETE FROM pets WHERE adopted = 0")
47
```

- Function name - snake-case naming styling
- No exception type was mentioned before
- Using method which difficult to guess adoption code



```
40  - def shelterStatus():
41  -     if adoption_count > 0: return "active"
42  -     return "quiet"
43
44
45    def main():
46  -     registerPet("Bruno", "dog", 7)
47  -     registerPet("Mitzi", "cat", 3)
48  -     registerPet("Kiwi", "parrot", 12)
49        adopt(1)
50  -     print(oldestAvailable(), adoptionCode(), shelterStatus())
51  -     clearUnadopted()
52
53
54  - main()
```

```
49  + def shelter_status():
50  +     """Return whether the shelter has recorded any adoptions."""
51  +     adopted = conn.execute(
52  +         "SELECT COUNT(*) FROM pets WHERE adopted = 1"
53  +     ).fetchone()[0]
54  +     return "active" if adopted else "quiet"
55
56
57    def main():
58  +     """Run the shelter demonstration."""
59  +     register_pet("Bruno", "dog", 7)
60  +     register_pet("Mitzi", "cat", 3)
61  +     register_pet("Kiwi", "parrot", 12)
62        adopt(1)
63  +     print(oldest_available(), adoption_code(), shelter_status())
64  +     clear_unadopted()
65
66
67  + if __name__ == "__main__":
68  +     main()
```

- Function name - snake-case naming style
- Checking adoption count from the database and not from the local running instance
- Not more than one statement in single line